Bounce-apart calculations

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use momentum and energy to check an elastic collision.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 6.*
Yes: sticking defines perfectly inelastic. Total momentum is conserved; kinetic energy need not be.

02. Worked example: explain the model
Before +3; after -1+4=+3 kg·m/s. The carts have different masses, so momentum-arrow lengths differ from velocity-arrow lengths.

03. Read and interpret the model
A stops; B moves right at 3 m/s.

04. Apply the physics
p_i = +5 kg·m/s; p_f = -1 kg·m/s. Δp = -1 - 5 = -6 kg·m/s, directed left.

05. Explain your reasoning
No. It is inelastic; momentum conservation alone is insufficient.

06. Review: connect an earlier idea *Review: Unit 2, Day 13.*
Velocity tangent; acceleration inward at both positions. Constant speed does not mean constant velocity.

07. Exit check
A-1 m/s; B+4 m/s.

08. STRETCH: test the model
Initial momentum 6; heavy final 2; light final momentum 4, so 4 m/s right.

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Learning target: Use momentum and energy to check an elastic collision.

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01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 6.*
A stops; B moves at +5 m/s. Equal masses exchange velocities in an ideal 1 D elastic collision.

02. Worked example: explain the model
Before +3; after -1+4=+3 kg·m/s. The carts have different masses, so momentum-arrow lengths differ from velocity-arrow lengths.

03. Read and interpret the model
A stops; B moves right at 3 m/s.

04. Apply the physics
4=-2+3 vB; vB 2 m/s. KE before 8 J; after 2+6=8 J.

05. Explain your reasoning
No. It is inelastic; momentum conservation alone is insufficient.

06. Review: connect an earlier idea *Review: Unit 2, Day 13.*
Speed becomes 1/4; period becomes 64 times as large. v∝r^(-1/2), T∝r^(3/2).

07. Exit check
vf 1 m/s; KE before 5 J, after 1 J. They stick and lose 4 J kinetic energy.

08. STRETCH: test the model
Initial total 7; -4+3 v=7 gives v=11/3 m/s. Impulses -14 and +14 N·s.